

Perturbative Weights for the 3×3 Model

$$H = \begin{pmatrix} -A & V & 0 \\ V & -9A & V \\ 0 & V & -25A \end{pmatrix}$$

1D analytical derivations

1 Motivation

We want the mod-square projections $|\langle \phi_k | \psi_n \rangle|^2$ of the eigenvectors of

$$H = \begin{pmatrix} -A & V & 0 \\ V & -9A & V \\ 0 & V & -25A \end{pmatrix} \quad (1)$$

onto the standard basis $\{|\phi_1\rangle, |\phi_2\rangle, |\phi_3\rangle\}$, expanded as Taylor series in V around $V = 0$.

A direct call to **Eigenvectors** on H in *Mathematica* produces **Root** objects (the characteristic polynomial is cubic) and, after **Normalize**, expressions involving **Abs**[...] and branch cuts. The function **Series** on these objects emits **Root::sbr** warnings and produces unwieldy output. We avoid the cubic by using non-degenerate Rayleigh–Schrödinger perturbation theory directly.

2 Splitting the Hamiltonian

Write

$$H = H_0 + W, \quad H_0 = \begin{pmatrix} -A & 0 & 0 \\ 0 & -9A & 0 \\ 0 & 0 & -25A \end{pmatrix}, \quad W = V \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}. \quad (2)$$

The unperturbed basis is the standard basis, with energies

$$E_1^{(0)} = -A, \quad E_2^{(0)} = -9A, \quad E_3^{(0)} = -25A. \quad (3)$$

All three are distinct, so standard non-degenerate perturbation theory applies. The convergence requirement is $V \ll A$.

3 Expansion and matching orders

For each band n expand

$$|\psi_n\rangle = |\phi_n\rangle + |\psi_n^{(1)}\rangle + |\psi_n^{(2)}\rangle + \cdots, \quad E_n = E_n^{(0)} + E_n^{(1)} + E_n^{(2)} + \cdots, \quad (4)$$

with $|\psi_n^{(k)}\rangle = \mathcal{O}(V^k)$. Substituting into $(H_0 + W)|\psi_n\rangle = E_n|\psi_n\rangle$ and matching powers of V :

$$\mathcal{O}(V^0) : H_0|\phi_n\rangle = E_n^{(0)}|\phi_n\rangle, \quad (5)$$

$$\mathcal{O}(V^1) : H_0|\psi_n^{(1)}\rangle + W|\phi_n\rangle = E_n^{(1)}|\phi_n\rangle + E_n^{(0)}|\psi_n^{(1)}\rangle, \quad (6)$$

$$\mathcal{O}(V^2) : H_0|\psi_n^{(2)}\rangle + W|\psi_n^{(1)}\rangle = E_n^{(2)}|\phi_n\rangle + E_n^{(1)}|\psi_n^{(1)}\rangle + E_n^{(0)}|\psi_n^{(2)}\rangle. \quad (7)$$

4 First-order corrections

Project Eq. (6) on $\langle\phi_n|$; the two H_0 terms cancel by Eq. (5), leaving

$$E_n^{(1)} = \langle\phi_n|W|\phi_n\rangle = V M_{nn}. \quad (8)$$

Since the diagonal of M is zero, $E_n^{(1)} = 0$ for all n : the off-diagonal perturbation does not shift the energies at first order.

Project Eq. (6) on $\langle\phi_m|$ with $m \neq n$ (so $E_n^{(1)}\langle\phi_m|\phi_n\rangle = 0$):

$$\langle\phi_m|\psi_n^{(1)}\rangle = \frac{W_{mn}}{E_n^{(0)} - E_m^{(0)}} = \frac{V M_{mn}}{E_n^{(0)} - E_m^{(0)}}. \quad (9)$$

For our matrix, only $M_{12} = M_{21} = M_{23} = M_{32} = 1$ are nonzero:

$$|\psi_1\rangle \approx |1\rangle + \frac{V}{8A}|2\rangle, \quad (10)$$

$$|\psi_2\rangle \approx |2\rangle - \frac{V}{8A}|1\rangle + \frac{V}{16A}|3\rangle, \quad (11)$$

$$|\psi_3\rangle \approx |3\rangle - \frac{V}{16A}|2\rangle. \quad (12)$$

5 Weights to $\mathcal{O}(V^2)$

The weight of eigenstate $|\psi_n\rangle$ on basis state $|\phi_k\rangle$ is $W_{kn}(V) = |\langle\phi_k|\psi_n\rangle|^2$.

Off-diagonal ($k \neq n$). From Eq. (9), the leading contribution comes from the first-order correction:

$$W_{kn}(V) = \frac{V^2 M_{kn}^2}{(E_n^{(0)} - E_k^{(0)})^2} + \mathcal{O}(V^4). \quad (13)$$

Diagonal ($k = n$). The wavefunction is not normalized at first order; expanding $|\tilde{\psi}_n\rangle = (1 - \frac{1}{2}\langle\psi_n^{(1)}|\psi_n^{(1)}\rangle)(|\phi_n\rangle + |\psi_n^{(1)}\rangle) + \mathcal{O}(V^2)$,

$$W_{nn}(V) = 1 - V^2 \sum_{m \neq n} \frac{M_{mn}^2}{(E_n^{(0)} - E_m^{(0)})^2} + \mathcal{O}(V^4). \quad (14)$$

This is probability conservation to second order: weight leaving $|\phi_n\rangle$ reappears on the coupled states.

6 Explicit weights for the model Hamiltonian

The nonzero matrix elements are $M_{12} = M_{23} = 1$, the gaps are $|E_1^{(0)} - E_2^{(0)}| = 8A$, $|E_2^{(0)} - E_3^{(0)}| = 16A$, $|E_1^{(0)} - E_3^{(0)}| = 24A$.

6.1 Weights on the upper band $|\phi_1\rangle$ ($k = 1$)

$$W_{11}(V) = 1 - \frac{V^2}{(8A)^2} + \mathcal{O}(V^4) = 1 - \frac{V^2}{64A^2} + \mathcal{O}(V^4), \quad (15)$$

$$W_{21}(V) = \frac{V^2}{(8A)^2} + \mathcal{O}(V^4) = \frac{V^2}{64A^2} + \mathcal{O}(V^4), \quad (16)$$

$$W_{31}(V) = \frac{V^4}{(8A \cdot 16A)^2} + \mathcal{O}(V^6) = \frac{V^4}{147456 A^4} + \mathcal{O}(V^6). \quad (17)$$

The third row starts at V^4 because $M_{13} = 0$; it comes from a second-order virtual transition $|3\rangle \rightarrow |2\rangle \rightarrow |1\rangle$ with amplitude $V/(16A) \cdot V/(8A) = V^2/(128A^2)$.

6.2 Weights on the middle band $|\phi_2\rangle$ ($k = 2$)

$$W_{12}(V) = \frac{V^2}{(8A)^2} + \mathcal{O}(V^4) = \frac{V^2}{64A^2} + \mathcal{O}(V^4), \quad (18)$$

$$W_{22}(V) = 1 - \frac{V^2}{(8A)^2} - \frac{V^2}{(16A)^2} + \mathcal{O}(V^4) = 1 - \frac{V^2}{64A^2} - \frac{V^2}{256A^2} + \mathcal{O}(V^4), \quad (19)$$

$$W_{32}(V) = \frac{V^2}{(16A)^2} + \mathcal{O}(V^4) = \frac{V^2}{256A^2} + \mathcal{O}(V^4). \quad (20)$$

6.3 Weights on the lower band $|\phi_3\rangle$ ($k = 3$)

$$W_{13}(V) = \frac{V^4}{(8A \cdot 16A)^2} + \mathcal{O}(V^6) = \frac{V^4}{147456 A^4} + \mathcal{O}(V^6), \quad (21)$$

$$W_{23}(V) = \frac{V^2}{(16A)^2} + \mathcal{O}(V^4) = \frac{V^2}{256A^2} + \mathcal{O}(V^4), \quad (22)$$

$$W_{33}(V) = 1 - \frac{V^2}{(16A)^2} + \mathcal{O}(V^4) = 1 - \frac{V^2}{256A^2} + \mathcal{O}(V^4). \quad (23)$$

6.4 Closure check

Each column sums to $1 + \mathcal{O}(V^4)$:

- $W_{11} + W_{21} + W_{31} = 1 - V^2/(64A^2) + V^2/(64A^2) + \mathcal{O}(V^4) = 1 + \mathcal{O}(V^4),$
- $W_{12} + W_{22} + W_{32} = V^2/(64A^2) + 1 - V^2/(64A^2) - V^2/(256A^2) + V^2/(256A^2) = 1$ (*exact to $\mathcal{O}(V^4)$*),
- $W_{13} + W_{23} + W_{33} = \mathcal{O}(V^4) + V^2/(256A^2) + 1 - V^2/(256A^2) + \mathcal{O}(V^4) = 1 + \mathcal{O}(V^4).$

Note that the V^2 weights leaking out of $|\phi_2\rangle$ split equally between $|\phi_1\rangle$ and $|\phi_3\rangle$ because $|\phi_2\rangle$ couples symmetrically to both.

7 Mathematica implementation

The cleanest transcription is one short block:

```
Clear[A, V];
H0 = DiagonalMatrix[{-A, -9 A, -25 A}];
M = {{0, 1, 0}, {1, 0, 1}, {0, 1, 0}};

(* first-order coefficients <phi_m|psi_n^(1)> *)
c[n_, m_] := V M[[n, m]]/(H0[[n, n]] - H0[[m, m]])

(* weight on |phi_k> from eigenstate |psi_n>, to O(V^2) *)
W[n_, k_] := Piecewise[{
  {1 - V^2 * Sum[If[m != n,
    M[[n, m]]^2/(H0[[n, n]] - H0[[m, m]])^2, 0], {m, 3}},
  {V^2 * M[[n, k]]^2/(H0[[n, n]] - H0[[k, k]])^2, k != n}
}] + O[V]^7

TableForm[Table[W[n, k], {n, 3}, {k, 3}]]
```

The output reproduces the closed-form expressions above, with no `Root`, no `Abs`, and no branch-cut warnings.

8 Weight ratio at the energy of band 2

We want the ratio

$$R(V) = \frac{|\langle \phi_1 | \psi_2 \rangle|^2}{|\langle \phi_2 | \psi_2 \rangle|^2} = \frac{W_{12}(V)}{W_{22}(V)}, \quad (24)$$

i.e. the spectral weights of the *unperturbed* bands 1 and 2 inside the band-2 eigenstate at $k = 1.5G$ (where the diagonal energies are $-A, -9A, -25A$).

8.1 Setup

At $k = 1.5G$ the unperturbed energies are $E_1^{(0)} = -A$, $E_2^{(0)} = -9A$, $E_3^{(0)} = -25A$, all distinct. The non-degenerate Rayleigh–Schrödinger machinery from Sec. 13–14 applies.

8.2 First-order wavefunction at $k = 1.5G$

Apply Eq. (9) with $n = 2$ and the only nonzero perturbation matrix elements $M_{21} = M_{23} = 1$:

$$\langle \phi_1 | \psi_2^{(1)} \rangle = \frac{V}{E_2^{(0)} - E_1^{(0)}} = \frac{V}{-9A - (-A)} = -\frac{V}{8A}, \quad (25)$$

$$\langle \phi_3 | \psi_2^{(1)} \rangle = \frac{V}{E_2^{(0)} - E_3^{(0)}} = \frac{V}{-9A - (-25A)} = +\frac{V}{16A}. \quad (26)$$

So the band-2 eigenstate to first order is

$$|\psi_2\rangle \approx |2\rangle - \frac{V}{8A}|1\rangle + \frac{V}{16A}|3\rangle. \quad (27)$$

8.3 Weights to $\mathcal{O}(V^2)$

Apply Eqs. (13)–(14). The off-diagonal weight of band 1 on $|\psi_2\rangle$ is

$$W_{12}(V) = \frac{V^2}{(8A)^2} + \mathcal{O}(V^4) = \frac{V^2}{64A^2} + \mathcal{O}(V^4), \quad (28)$$

and the diagonal weight of band 2 is

$$W_{22}(V) = 1 - \frac{V^2}{(8A)^2} - \frac{V^2}{(16A)^2} + \mathcal{O}(V^4) = 1 - \frac{V^2}{64A^2} - \frac{V^2}{256A^2} + \mathcal{O}(V^4). \quad (29)$$

8.4 Analytical weight ratio

Taking the ratio and keeping terms up to $\mathcal{O}(V^4)$:

$$\begin{aligned} R(V) &= \frac{W_{12}(V)}{W_{22}(V)} = \frac{V^2/(64A^2)}{1 - V^2/(64A^2) - V^2/(256A^2)} + \mathcal{O}(V^6) \\ &= \frac{V^2}{64A^2} \cdot \frac{1}{1 - 5V^2/(256A^2)} + \mathcal{O}(V^6) \\ &= \boxed{\frac{V^2}{64A^2} + \frac{5V^4}{16384A^4} + \mathcal{O}(V^6)}. \end{aligned} \quad (30)$$

The key observation: the normalization factor $1 - 5V^2/(256A^2)$ in the denominator of W_{22} *cancels* in the ratio, leaving the leading term $V^2/(64A^2)$ as the clean analytical result.

8.5 Mathematica implementation of the ratio

The ratio $R(V)$ can be computed three ways in *Mathematica*: from the perturbative formulas, as a Taylor series, and exactly via the cubic. Here is a single self-contained cell that does all three and cross-checks them:

```
Clear[A, V];
H0 = DiagonalMatrix[{-A, -9 A, -25 A}];
M = {{0, 1, 0}, {1, 0, 1}, {0, 1, 0}};

(* ---- (1) perturbative weights (reusing W[n,k] from earlier) ---- *)
c[n_, m_] := V M[[n, m]]/(H0[[n, n]] - H0[[m, m]])
W[n_, k_] := Piecewise[{
  {1 - V^2 * Sum[If[m != n,
    M[[n, m]]^2/(H0[[n, n]] - H0[[m, m]])^2, 0], {m, 3}},
  k == n},
  {V^2 * M[[n, k]]^2/(H0[[n, n]] - H0[[k, k]])^2, k != n}
] + O[V]^7

Rpert[V_] := (W[2, 1]/W[2, 2]) // Normal // Simplify[#, A > 0] &

(* ---- (2) perturbative Taylor expansion of the ratio ---- *)
Rtaylor[V_, nMax_] := Series[W[2, 1]/W[2, 2], {V, 0, nMax},
```

```

Assumptions -> A > 0] // Normal // Simplify

(* ---- (3) exact ratio via the cubic eigenvalue ---- *)
H[kG_: 1.5] := DiagonalMatrix[{-A (kG - 1)^2 *4, -A kG^2 *4, -A (kG + 1)^2 *4}]
               + V {{0, 1, 0}, {1, 0, 1}, {0, 1, 0}};
Rexact[V_] := Module[{E2, Hmat},
  Hmat = H[1.5];
  E2 = Root[CharacteristicPolynomial[Hmat, \[Lambda]] &, 2]; (* middle root *)
  V^2/(A + E2)^2 // Simplify
]

(* ---- cross-check ---- *)
Block[{Vset = 0.3, Aval = 1.0},
  Print["Perturbative : ", Rpert[Vset] /. A -> Aval];
  Print["Taylor (V^6) : ", Rtaylor[Vset, 6] /. A -> Aval];
  Print["Exact cubic : ", Rexact[Vset] /. A -> Aval];
  Print["Leading form : ", (Vset^2/(64 Aval^2))];
]

```

Expected output (with $A = 1$, $V = 0.3$):

```

Perturbative : 0.00140428
Taylor (V^6) : 0.00140428
Exact cubic   : 0.00140428
Leading form  : 0.00140625

```

The first three agree to five digits because the perturbation series is accurate to $\mathcal{O}(V^4)$; the leading-order approximation $V^2/(64A^2) = 1.40625 \times 10^{-3}$ is off by $\sim 0.1\%$, exactly the size of the $5V^4/(16384A^4) = 2.47 \times 10^{-6}$ correction.

Plot of $R(V)$ vs V :

```

Plot[{Rtaylor[v, 6], Rexact[v]}, {v, 0, 1.5},
  PlotLegends -> {"perturbative  $\mathcal{O}(V^6)$ ", "exact cubic"},
  AxesLabel -> {"V/A", "W12/W22"},
  Epilog -> {Dashed, Line[{0, 0}, {1.5, (1.5)^2/64}]}]}

```

The dashed line is the leading-order $V^2/(64A^2)$; the curves peel off from it as V grows.

8.6 Exact expression via the cubic eigenvalue

If one insists on the exact answer (not perturbative), solve $(H - E_2 I)|\psi_2\rangle = 0$. The middle-band eigenvalue satisfies the cubic

$$E_2^3 + 35A E_2^2 + (259A^2 - 2V^2)E_2 + (225A^3 - 26AV^2) = 0, \quad (31)$$

and the eigenvector components satisfy

$$\alpha = \frac{V\beta}{A + E_2}, \quad \gamma = \frac{V\beta}{25A + E_2}, \quad (32)$$

with β fixed by normalization. Therefore

$$\left. \frac{W_{12}}{W_{22}} \right|_{\text{exact}} = \frac{V^2}{(A + E_2)^2}, \quad (33)$$

where E_2 is the middle root of the cubic above. This reduces to $V^2/(64A^2)$ for $V \rightarrow 0$ since $E_2 \rightarrow -9A$ and $A + E_2 \rightarrow -8A$.

8.7 General k : where does the formula come from?

For general k , the unperturbed energies $\epsilon_n(k) = -\alpha(k + nG)^2$ enter only through the *gaps*

$$\Delta_{21}(k) = \epsilon_2(k) - \epsilon_1(k), \quad \Delta_{23}(k) = \epsilon_2(k) - \epsilon_3(k), \quad (34)$$

and the weight ratio becomes

$$\frac{W_{12}(k)}{W_{22}(k)} = \frac{V^2/\Delta_{21}(k)^2}{1 - V^2/\Delta_{21}(k)^2 - V^2/\Delta_{23}(k)^2} + \mathcal{O}(V^6). \quad (35)$$

At $k = 1.5G$: $\Delta_{21} = -8A$ and $\Delta_{23} = 16A$, giving the result derived above. At $k = -1.5G$ the matrix is index-reversed (diagonals $-25A, -9A, -A$), but the middle band is still $|2\rangle$ with the same gap structure, so the ratio is unchanged.

8.8 Physical interpretation

The factor $V^2/(64A^2) = (V/\Delta_{21})^2$ is the squared mixing amplitude between bands 1 and 2 from first-order perturbation theory: the perturbation V couples the unperturbed states with amplitude V/Δ , and the weight is the modulus squared. The gap $\Delta_{21} = 8A$ between bands 1 and 2 is the *only* scale entering the leading-order result – this is the standard textbook picture of band repulsion: closer bands mix more.

9 Why this approach wins for this matrix

- **No cubic to solve.** Every coefficient is a rational function of A , immediately readable.
- **No branch cuts.** The result is a polynomial in V^2 with rational coefficients in A .
- **Physically transparent.** The amplitude $V/(E_n^{(0)} - E_m^{(0)})$ is the textbook mixing: larger gaps $\Rightarrow$ smaller mixing.
- **Systematic extension.** Going to higher order requires one more projection at each step; no new machinery.
- **Convergence.** Valid for $V/A \ll 1$; the band closest in energy ($|\phi_2\rangle$ to $|\phi_1\rangle$, gap $8A$) sets the strictest bound.